Machine Learning-Based Phishing Website Detection

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Step 1: Dataset Selection

Dataset Name: Web Page Phishing Dataset

Kaggle URL: `https://www.kaggle.com/datasets/danielfernandon/web-page-phishing-dataset`

Dataset Description:

I selected the Web Page Phishing Dataset to investigate how machine learning can identify phishing websites. The dataset combines two existing phishing datasets and contains numerical characteristics of website URLs that may help distinguish legitimate websites from phishing websites.

Number of Records: 100,077

Number of Features: 19 input features and one target column, for a total of 20 columns.

Target Variable: phishing

The target variable identifies whether a website is legitimate (0) or phishing (1). The dataset contains 63,715 legitimate records and 36,362 phishing records.

Important Features:

- URL Length: Total length of the URL.
- Number of Dots: Counts dots in the URL.
- Number of Hyphens: Counts hyphens in the URL.
- Number of Slashes: Counts slash characters.
- Number of Question Marks: Counts question marks.
- Number of @ Symbols: Counts @ characters.
- Number of Redirections: Records URL redirection characteristics.

These features will be analyzed to determine whether URL characteristics can help machine learning models detect phishing websites.

Step 2: Define a Security Problem

Security Problem:

Phishing websites are a common cybersecurity threat because they are designed to look like legitimate websites. Attackers use these fake websites to trick users into sharing sensitive information, such as passwords, personal details, and financial information. Since phishing websites can be difficult to recognize, machine learning may help identify suspicious patterns in website URLs.

Research Question:

Can Decision Tree and Random Forest machine learning models accurately distinguish phishing websites from legitimate websites using URL characteristics?

Project Objective:

The objective of this project is to analyze website URL characteristics and compare two machine learning models, Decision Tree and Random Forest, to determine how effectively they can detect phishing websites. The project will also examine false positives and false negatives to understand the security risks of incorrect predictions.

Step 3: Data Exploration

Dataset Overview

The Web Page Phishing Dataset originally contained 100,077 records and 20 columns. Of these columns, 19 represent URL characteristics, while the phishing column is the target variable.

Class Distribution

The original dataset contained 63,715 legitimate websites and 36,362 phishing websites.

Missing Values and Duplicates

The dataset contained no missing values. However, 78,186 exact duplicate records were identified. After removing duplicates, 21,891 records remained.

Feature Analysis

The analysis showed that phishing websites generally had longer URLs and more slashes than legitimate websites.

Correlation analysis also showed that the number of slashes was strongly associated with the phishing classification.

Website Class Distribution

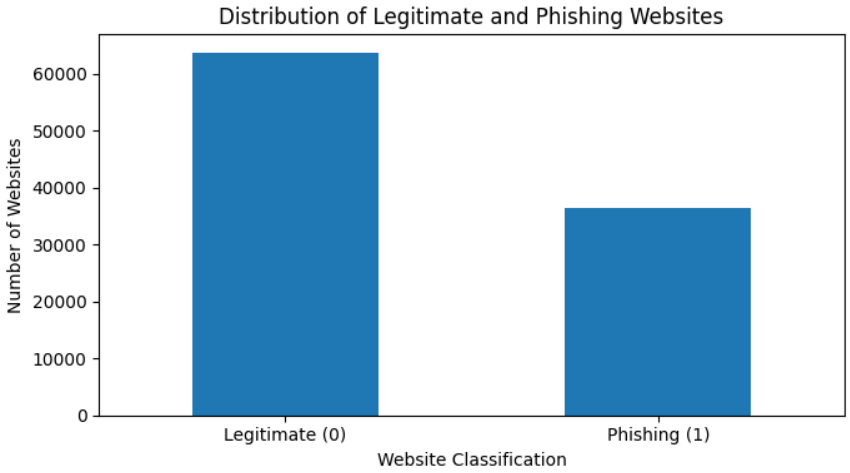


Figure 1: Distribution of legitimate and phishing websites

Average URL Length

The analysis showed that phishing websites had an average URL length of 66.49 characters, compared to 23.59 characters for legitimate websites. This suggests that URL length may be a useful feature for phishing detection.

Average Number of Slashes

The analysis showed that phishing websites contained an average of 2.62 slashes, compared to 0.29 slashes for legitimate websites. This suggests that the number of slashes may help distinguish phishing websites from legitimate websites.

Correlation Analysis

The correlation analysis showed that the number of slashes had the strongest positive correlation with phishing classification (0.611), followed by URL length (0.430) and the number of equal signs (0.260).

These findings suggest that URL structure provides useful information for identifying potential phishing websites.

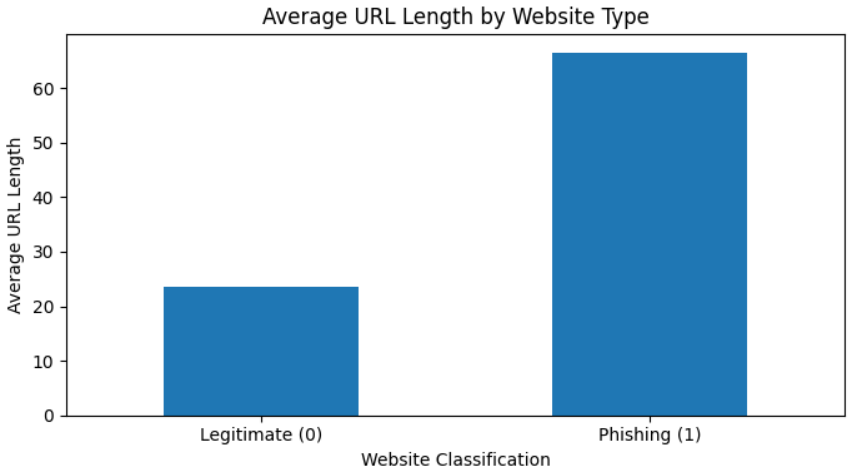


Figure 2: Average URL length of legitimate and phishing websites

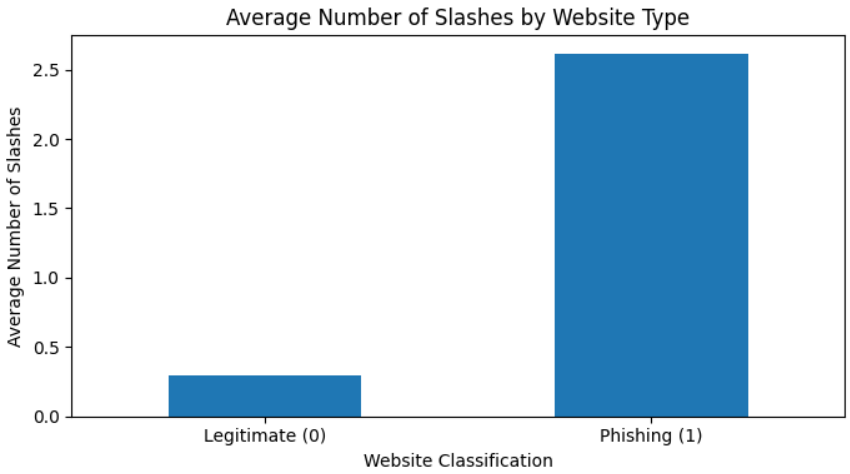


Figure 3: Average number of slashes in legitimate and phishing URLs

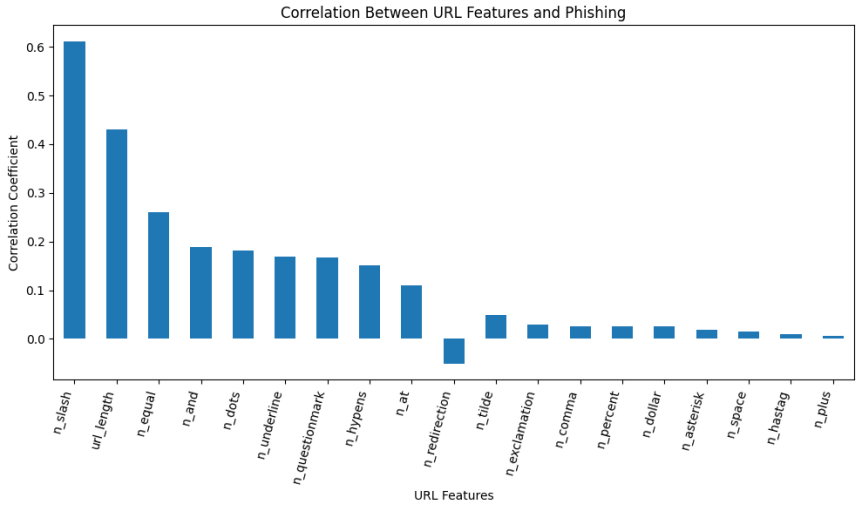


Figure 4: Correlation between URL characteristics and phishing classification

Step 4: Data Preprocessing

Handling Missing Values

The dataset was checked for missing values. No missing values were found, so no additional processing was required.

Removing Duplicate Records

The original dataset contained 78,186 exact duplicate records. These were removed to reduce repetition and prevent duplicate records from affecting model evaluation.

After cleaning, 21,891 unique records remained.

Feature Preparation

The dataset contained 19 numerical URL features used to predict the phishing classification. Categorical encoding was not required.

Feature scaling was not necessary because Decision Tree and Random Forest models do not require standardized feature values.

Train and Test Split

The cleaned dataset was divided into training and testing sets.

The training set contained 17,523 records, while the testing set contained 4,368 records.

A group-aware split was used to prevent identical feature combinations from appearing in both the training and testing sets.

Class Imbalance

After removing duplicates, the dataset contained 15,872 phishing records and 6,019 legitimate records.

The class imbalance was considered when evaluating model performance, especially precision, recall, and false predictions.

Step 5: Machine Learning Methods

Decision Tree

The Decision Tree classifier was used to predict whether a website was legitimate or phishing based on its URL characteristics.

This model makes predictions by learning decision rules from the training data.

Random Forest

The Random Forest classifier was also trained to detect phishing websites.

Random Forest combines multiple decision trees to produce predictions. This can improve performance compared to using a single Decision Tree.

Model Training

Both models were trained using the same 17,523 training records and evaluated using the same 4,368 testing records.

The models used 19 numerical URL features to predict the phishing classification.

Purpose of Comparison

The purpose of comparing these two models was to determine which approach could more accurately identify phishing websites while minimizing incorrect predictions.

Step 6: Model Evaluation

Model Accuracy

The Decision Tree model achieved an accuracy of 81.41%, while the Random Forest model achieved an accuracy of 82.58%.

Random Forest performed slightly better overall in identifying legitimate and phishing websites.

Random Forest Classification Metrics

For the phishing class, the Random Forest model achieved a precision of approximately 84%, recall of 94%, and F1-score of 89%.

The high recall indicates that the model successfully identified most phishing websites in the test dataset.

However, its recall for legitimate websites was approximately 50%, indicating that many legitimate websites were incorrectly flagged as phishing.

Confusion Matrix Analysis

Confusion matrices were used to evaluate how accurately each model classified legitimate and phishing websites.

The Decision Tree correctly identified 2,874 phishing websites but missed 351 phishing websites. It also incorrectly classified 461 legitimate websites as phishing.

The Random Forest correctly identified 3,034 phishing websites but missed 191 phishing websites. It incorrectly classified 570 legitimate websites as phishing.

Security Implications

False positives occur when legitimate websites are incorrectly identified as phishing. This may prevent users from accessing safe websites.

False negatives occur when phishing websites are incorrectly identified as legitimate. These errors are especially dangerous because users may visit malicious websites without receiving a warning.

Although Random Forest produced more false positives, it detected more phishing websites and reduced the number of false negatives.

Decision Tree Confusion Matrix

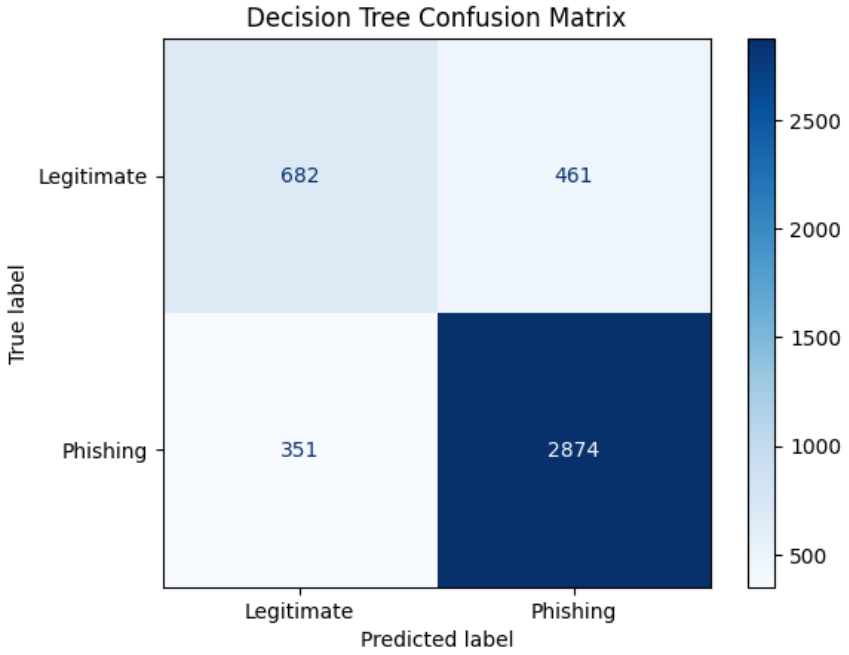


Figure 5: Decision Tree confusion matrix

Random Forest Confusion Matrix

Step 7: Security Interpretation

Phishing Detection

The machine learning models were able to identify phishing websites based on numerical URL characteristics.

Random Forest correctly detected 3,034 phishing websites, compared to 2,874 detected by Decision Tree.

False Positives and False Negatives

False positives may cause legitimate websites to be blocked or incorrectly flagged as unsafe.

False negatives are especially concerning because phishing websites may go undetected, potentially exposing users to credential theft or financial fraud.

Random Forest reduced false negatives from 351 to 191, although it increased false positives from 461 to 570.

Real-World Security Applications

These models could be used as part of a browser warning system or website security filter.

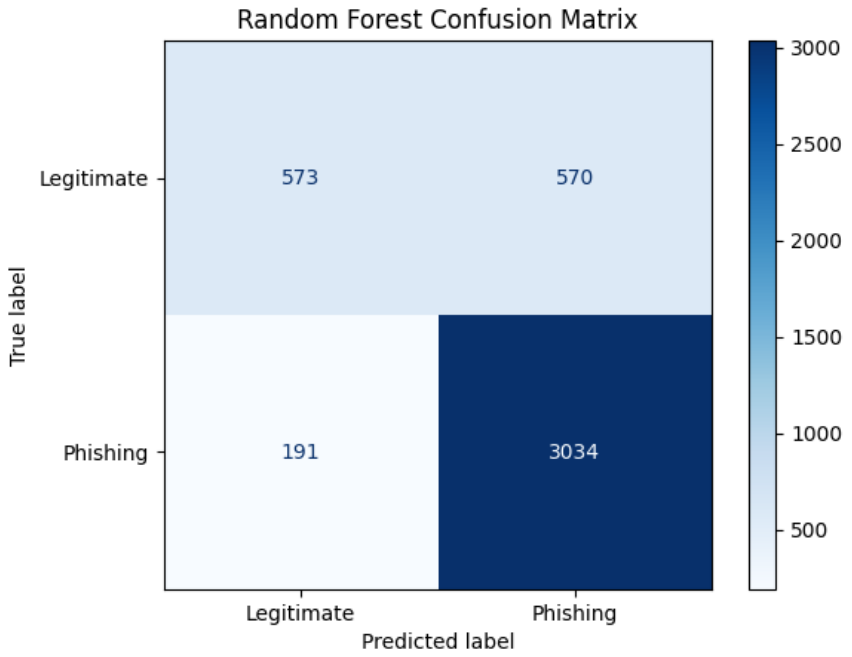


Figure 6: Random Forest confusion matrix

However, predictions should be combined with other security checks rather than being used as the only method of phishing detection.

Feature Importance

The Random Forest analysis identified the number of slashes (30.54%), URL length (19.41%), and number of hyphens (13.58%) as the three most important features.

These results suggest that URL structure can help machine learning models identify potential phishing websites.

Random Forest Feature Importance

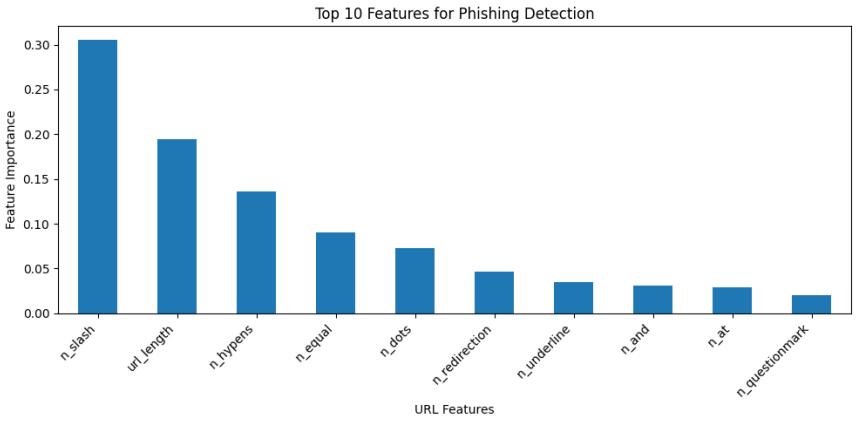


Figure 7: Top 10 most important URL features in Random Forest

Step 8: Limitations

One limitation of this project is that the models rely only on numerical URL characteristics rather than analyzing website content or user behavior.

The original dataset contained many duplicate records, and some identical URL feature combinations had conflicting classification labels. These issues may affect the reliability of predictions.

Additionally, the cleaned dataset contained more phishing websites than legitimate websites, creating a class imbalance.

The models were evaluated using one dataset, so their performance on new websites may differ from the reported results.

Step 9: Conclusion

This project investigated whether machine learning can detect phishing websites using URL characteristics.

Two classification models were evaluated: Decision Tree and Random Forest.

Decision Tree achieved 81.41% accuracy, while Random Forest achieved 82.58%.

Random Forest detected more phishing websites and produced fewer false negatives, although it also generated more false positives.

The findings demonstrate that machine learning can support phishing detection, but it should be combined with additional security measures for more reliable protection.

Step 10: References

1. Web Page Phishing Dataset. Kaggle. `https://www.kaggle.com/datasets/danielfernandon/web-page-phishing-data`
2. Scikit-learn Documentation. `https://scikit-learn.org/stable/`
3. Pandas Documentation. `https://pandas.pydata.org/docs/`